Multi-task Learning for Speech Emotion and Emotion Intensity Recognition

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***Abstract*—Speech emotion recognition (SER) helps achieve better human-computer interaction and thus has attracted extensive attention from industry and academia. Speech emotion intensity plays an important role in the emotional description, but its effect on emotion recognition still has been rarely studied in the area of SER to the best of our knowledge. Previous studies have shown that there is a certain relationship between speech emotion intensity and emotion category, so each recognition task of multi-task learning is supposed to be beneficial to each other. We propose a multi-task learning framework with a self-supervised speech representation extractor based on Wav2Vec 2.0 to detect speech emotion and intensity at the same time in downstream networks. Experiment results show that the multi-task learning framework outperforms SOTA SER models and achieves 5% and 7% SER performance improvement on IEMOCAP and RAVDESS thanks to the auxiliary task of emotion intensity recognition.**

***Index Terms*—Speech emotion recognition, Multi-task learning, Speech emotion intensity**

I. Introduction

Speech emotion recognition (SER) helps machines behave and communicate in a more humanlike way and becomes an essential research branch of artificial intelligence [1][2][3]. Besides, SER has wide application perspectives on emergency discovery, fraud detection, psychological comfort, etc. However, the accuracy of SER still needs improvement, of which one reason is the poor speech emotion representation ability.

Emotion intensity or arousal is closely related to emotion category or emotion valence. [4] indicated that high-intensity speech clips had more negative valence which was true for both spontaneous and posed clips. Furthermore, typical "basic" emotions such as anger, fear, sadness, and disgust were most common amongst the high-intensity clips, whereas low-arousal emotions, such as boredom and contentment, were most frequent amongst the low-intensity clips. Besides, happiness was most common amongst the medium-intensity clips. [5] stated that each emotion may have different intensity level and detecting the intensity of an emotion can be beneficial for emotion analysis of text and speech.

To visualize the relationship between speech emotion intensity and emotion category, this paper presents the statistical correlation on the popular speech emotion dataset, namely IEMOCAP [6]. IEMOCAP provides 9 levels of arousal label which can be regarded as the intensity label according to previous studies of emotional arousal [7][8][9]. Fig. 1 shows the emotion intensity distribution on IEMOCAP, from which it can be observed that 1) The distribution of intensity conforms to Gaussian distribution for each of the emotions; 2) The proportion of anger and happy emotion increase with the increase of emotion intensity; 3) The proportion of sad emotion increases with the decrease of emotion intensity; 4) Neutral emotion are most common in medium-intensity speeches.

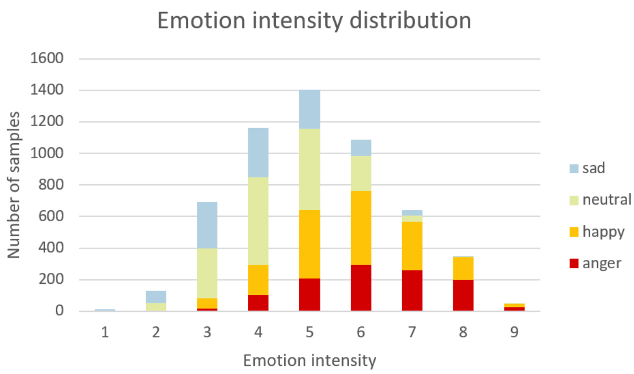

Fig. 1. Emotion intensity distribution of IEMOCAP

There were many works focused on emotion intensity in multimodal emotion researches. [10] introduced an effective interpolation technique that enabled better intensity control in TTS. [11] proposed a method to detect a player's emotions based on heart beat (HR) signals and facial expressions (FE). Besides, the emotion's intensity can be estimated by the HR value. [12] described a system developed for predicting emotion intensity associated with fear, anger, sadness, and joy in tweets. [13] proposed a multi-task ensemble framework that jointly learns multiple related problems including emotion and intensity classification on written text. However, speech emotion intensity recognition (SEIR) as a corresponding task of SER is seldomly studied. Inspired by the strong expressiveness of emotion intensity in other modals, we explore whether emotion intensity recognition of speech is helpful for SER. The statistical relationship between speech emotion and intensity is discussed above, which indicates that emotion intensity recognition is suitable as an auxiliary task of speech emotion recognition in the framework of multi-task learning (MTL).

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The intuition behind multi-task learning is that if two or more tasks are correlated, then the joint-model can learn effectively from the shared representations. In [14], the authors presented a multi-lingual and multi-task learning audio only SER system based on the multi-lingual pre-trained wav2vec 2.0 model, where gender and language classification along with mean and standard deviation of F0 and energy and voice ratio prediction tasks were considered as auxiliary tasks. [15] investigated training a single network to perform both automatic speech transcription and speaker recognition tasks jointly. [16] proposed to use MTL and use gender and naturalness as auxiliary tasks in deep neural networks in order to improve the generalization capability of the emotion models. They found that models using both gender and naturalness achieved more gains than those using either gender or naturalness separately. [17] built a multi-task learning model for the speaker verification and language identification tasks, to which the self-supervised framework Wav2Vec 2.0 was extended. [18] proposed a multi-task learning framework that used auxiliary tasks of gender identification and speaker recognition as auxiliary tasks with abundantly labelled data to improve the primary task of SER for which only limited data was available. Some work also discussed whether adversarial and multi-task learning was beneficial in speech recognition and came to the conclusion that deep models trained on big datasets already developed invariant representations to speakers in some cases [19]. Although so many auxiliary tasks were investigated for the benefit of SER primary task, emotion intensity recognition has not been considered to the best of our knowledge.

In this paper, we propose a multi-task learning framework to perform two tasks at the same time. The speech representation is extracted by the pre-trained model Wav2Vec 2.0 and different multi-layer perceptions are constructed to classify the emotion category and the emotion intensity. The contributions of this work are as follows: 1) This work is the first to propose a multi-task learning framework to perform speech emotion recognition as the primary task and intensity recognition as the auxiliary task; 2) The proposed framework requires only one unified model to recognize the speech emotion and the intensity simultaneously and provides richer emotional information than single task model; 3) The proposed framework achieves SOTA SER performance.

II. METHOD

Our framework of speech related multi-task learning consists of two major components: 1) A Speech Representation Extractor (SRE) based on Wav2Vec 2.0 which is trained in a self-supervised manner on raw audio data. 2) Downstream networks that operate on the extracted speech representations to classify the emotion and the emotion intensity. We first describe these two components and then present our multi-task training framework that trains the two downstream networks jointly with a common SRE backbone. The framework is presented in Fig. 2.

A. Wav2Vec 2.0 based speech representation extractor

For the speech representation extractor, we implement Vanilla fine-tuning Wav2Vec 2.0 reported in [20], which demonstrated the effectiveness of Wav2Vec 2.0 in the field of SER. The Wav2Vec 2.0 model extracts self-supervised speech representation by a multi-layer convolutional neural network taking raw speech signals as the input. Under the default setup in Wav2Vec 2.0 encoder, the CNN contains 7 layers of which the step sizes are (5,2,2,2,2,2,2) and the convolution kernel widths are (10,3,3,3,3,2,2). As the number of output channels of the last convolution in the encoder is set to 512, a 512-dimensional speech feature is generated for every 20ms speech. To utilize Wav2Vec representation to speech emotion recognition, Vanilla fine-tuning Wav2Vec 2.0 processes the final contextualized embedding generalized by Wav2Vec 2.0 using a global average pooling across the time dimension, and then classifies the emotion using the ReLU activation followed by a full connection layer. Concretely, the global average pooling unifies the length of the speech feature embedding extracted by the Wav2Vec 2.0 model to 768 regardless of the input speech length. Therefore, the speech representation of this extractor is desirable for the representation of a whole speech rather than speech fragments.

It should be noted that Wav2Vec 2.0 model is pre-trained with self-supervised training on a large number of corpora, thus the extracted speech representation of Wav2Vec 2.0 contains rich paralinguistic information such as speech emotion and emotion intensity, etc. Therefore, Wav2Vec 2.0 is desirable for extracting information needed for multi-task learning in this work.

B. Downstream networks

Downstream networks map the speech representation to task specific outputs. The two tasks in this work are speech emotion recognition and speech emotion intensity recognition. The emotion intensity recognition can be either a regression task or a classification task. For convenience and highlighting the emotion intensity in speech representation, we consider the speech emotion intensity recognition as a 2-class category task in this work. The two tasks in this work are similar as they both predict the classification of a complete speech according to the speech representation extracted from SRE. We construct similar fully connected layers to adapt to output dimensions corresponding to the number of classes of each downstream task.

C. Multi-task learning framework

In our multi-task learning framework, we train a shared SRE backbone with two downstream heads, which correspond to recognition of the speech emotion and emotion intensity.

We denote $representation_x$ as the speech representation of the input x extracted by the Wav2vec encoder. The fully connected layers that classify the speech emotion and intensity are presented as FC_{emo} and FC_{int} for short. Then the output of the downstream tasks are calculated as:

$$out_{emo} = FC_{emo}(representation_x),$$

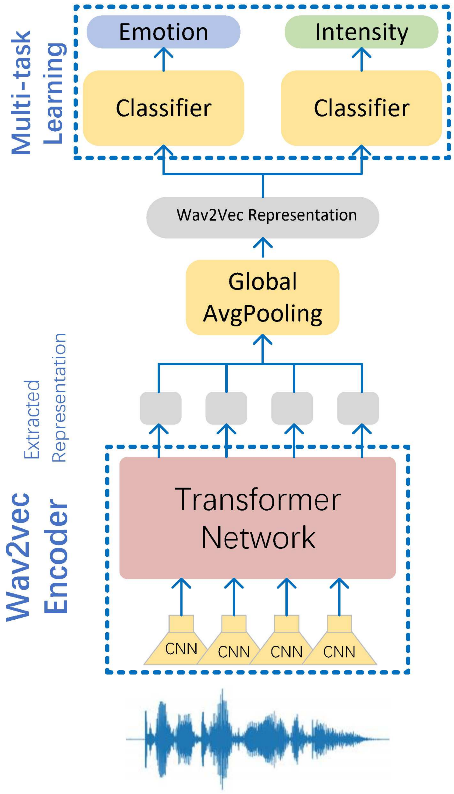


Fig. 2. Multi-task learning framework for speech emotion recognition as the primary task and emotion intensity recognition as the auxiliary task

and

$$out_{int} = FC_{int}(representation_x).$$

We denote l_{emo} and l_{int} as the ground truth label of emotion and intensity, respectively. Then the loss of the downstream tasks can be presented as: $Loss(out_{emo}, l_{emo})$ and $Loss(out_{int}, l_{int})$. In our multi-task learning framework, the training loss is the weighted sum of the loss of the downstream tasks. After a great number of experiments and tests, we empirically determine the weight and the multi-task learning loss calculation is as follows:

$$Loss_{MTL} = Loss(out_{emo}, l_{emo}) + 0.5 * Loss(out_{int}, l_{int}).$$

III. Experiment and results discussion

A. Datasets

Two databases in the field of speech emotion recognition, IEMOCAP and RAVDESS, are considered to prove the advance of our proposed method. The two datasets are described below.

IEMOCAP. Interactive Emotional Dyadic Motion Capture is a popular dataset for evaluating SER systems. It contains five recording sessions, each with one male speaker and one female speaker. Following the popular setup with previous SER works, we use the default labels provided by IEMOCAP, except that the “excited” category is merged with “happy” due to its sparsity in the dataset. Finally, only four emotion categories are considered: neutral, sad, angry, and happy. After that, the amounts of examples labeled ’neutral’, ’happy’, ’anger’, ’sad’ are 1708, 1636, 1103, 1084, respectively. The total amount of speech in IEMOCAP is about 7 hours. In this work, we only use the speech modality. In spite that there is no explicit emotion intensity label in IEMOCAP, each segment is annotated for the presence of valence, arousal and dominance, where arousal dimensional attribute can be considered as a label of emotion intensity. The original arousal attribute ranges from 1 to 5 with an interval of 0.5, which means the emotion intensity is divided into 9 levels.

RAVDESS. Ryerson Audio-Visual Database of Emotional Speech and Song dataset is a validated multi-modal database of emotional speech and song. This gender-balanced database consists of 24 professional actors, each performing 104 unique vocalizations with emotions that include: happy, sad, angry, fearful, surprise, disgust, calm, and neutral[21]. Each actor enacted 2 statements for each emotion: “Kids are talking by the door” and “Dogs are sitting by the door.” These statements were also recorded in two different emotion intensities, normal and strong, for each emotion, except for neutral (normal only). Actors repeated each vocalization twice. There are a total of 1440 speech utterances and 1012 song utterances. In this work, we only use the speech part.

B. Experimental setup

We cross validated the models on the two datasets. As for IEMOCAP, we split the dataset by leaving one session out as test set, and the remaining four sessions were used for training. As for RAVDESS, we split the dataset to six equal subsets according to the speaker ID, and chose one subset as test set while the rest subsets as training data. Under this setup we performed every experiment with multi fold cross validation. The experimental results presented the average value of cross validation. Four metrics were used to evaluate the performance, including unweighted accuracy rate (UAR), weighted accuracy rate (WAR), micro-F1, and macro-F1.

In the experiments of this work, we divided the emotion intensity of IEMOCAP into two categories despite that the arousal attribute had 9 levels. Limiting the categories of intensity classification is expected to help to extract the representation related to intensity since the training samples are very limited in IEMOCAP.

We compared the proposed multi-task learning framework which is abbreviated to $\text{MTL}_{emo+int}$ with single task learning which is represented as STL_{emo} for short. The performance comparison was carried out on two speech emotion datasets, namely IEMOCAP and RAVDESS, where labels describing the emotion intensity exist.

The common experimental hyperparameters in the finetuning stage are listed in Table I. Note that the finetuning used an exponentially decaying learning rate, which can be calculated as follows:

$$lr = lr_{init} * (r_{\text{decay}})^{\frac{Step_{\text{curr}}}{Step_{\text{decay}}}},$$

where lr_{init} denotes the initial learning rate, $Step_{\text{curr}}$ is the current number of training steps, and $Step_{\text{decay}}$ is the decay period.

TABLE I
COMMON HYPERPARAMETERS IN THE FINETUNING STAGE

Hyperparameters	Value
batch size	64
initial learning rate	0.0001
decay rate of learning rate	0.98
max speech length	6 (s)
max epochs	50
precision	32
optimizer	Adam

C. Experiment results and discussion

Firstly, the SER performance of multi-task and single task learning approaches are compared. Table II presents the speech emotion recognition results of the proposed multi-task framework on IEMOCAP, and Table III presents the results on RAVDESS. It can be observed that emotion intensity recognition task assisted speech emotion recognition of multi-task learning completely outperforms SER of single task learning. Specifically, the accuracy rate as well as F1 score of multi-task learning emotion recognition are 5% and 7% higher than that of single task learning in IEMOCAP and RAVDESS, respectively. The results demonstrate that emotion intensity recognition is a very effective auxiliary task for SER.

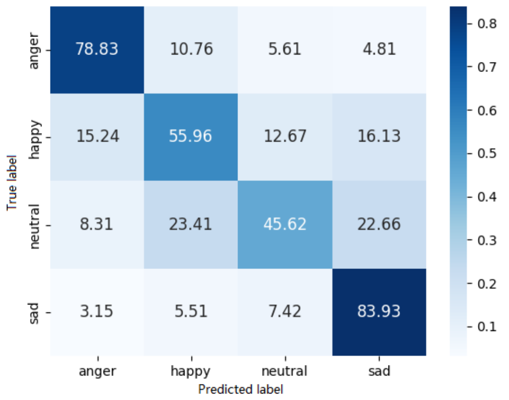
TABLE II
SPEECH EMOTION RECOGNITION RESULTS OF THE PROPOSED MULTI-TASK FRAMEWORK ON IEMOCAP

Model	UAR	WAR	micro-F1	macro-F1
STL_{emo}	0.6511	0.6268	0.6436	0.6245
$\text{MTL}_{emo+int}$	0.7082	0.6829	0.697	0.6864

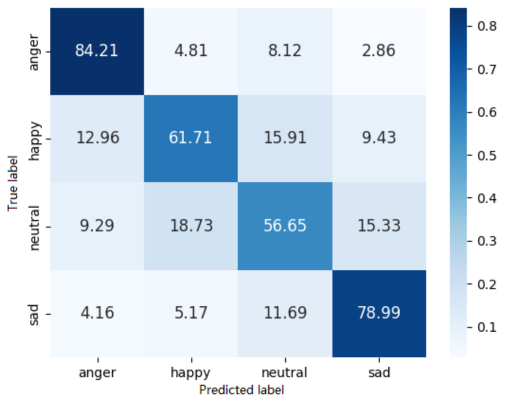
TABLE III
SPEECH EMOTION RECOGNITION RESULTS OF THE PROPOSED MULTI-TASK FRAMEWORK ON RAVDESS

Model	UAR	WAR	micro-F1	macro-F1
STL_{emo}	0.7237	0.7313	0.7311	0.7164
$\text{MTL}_{emo+int}$	0.8001	0.8007	0.8066	0.7996

Then SER confusion matrices are shown in Fig. 3 and Fig. 4 which provide more intuitive effect display of multi-task learning. From Fig. 3 it can be observed that the multi-task learning approach improves the recognition of emotion anger, happy and neutral, and the recognition accuracy of sad emotion decreased for some sad speeches are classified as neutral. Fig. 4 shows that on RAVDESS, the emotion recognition accuracy of most categories has been greatly improved by the multi-task learning approach, especially for emotion fearful and happy. The only category with decreased emotion recognition accuracy is neutral. Although the proposed approach does not gain performance improvement in all the emotion categories, the overall accuracy of SER is significantly improved by multi-task learning of speech emotion and intensity recognition.



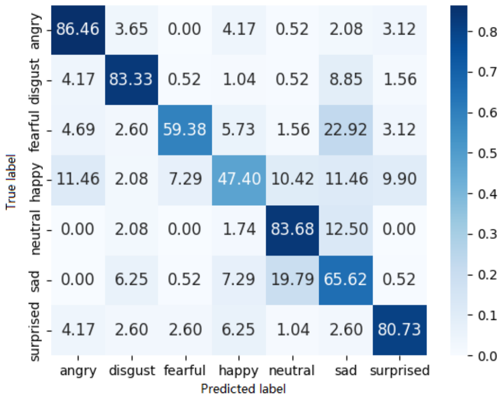
(a) single task learning for speech emotion recognition



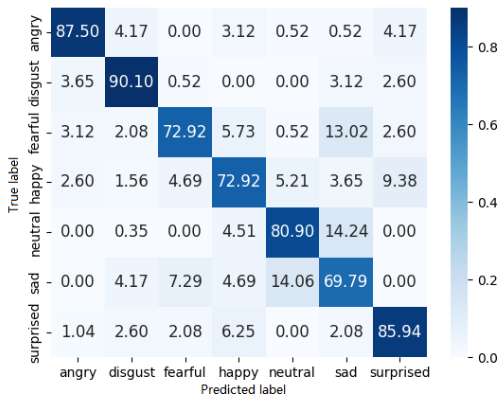
(b) multi-task learning for recognition of speech emotion and intensity

Fig. 3. SER confusion matrices of single task and multi-task learning approaches on IEMOCAP

Then we present the misclassification proportion of each emotion and intensity on IEMOCAP in Fig. 5 and Fig. 6. Note that we have divided the intensity into two categories to reduce the convergence problem caused by data scarcity. It can be observed from Fig. 5 that the emotion recognition error rate of weak intensity is quite high in emotion categories such as happy, angry, and neutral. This phenomenon coincides with the common sense that the emotion categories of speeches with low emotion intensity are often confused, except for sad emotion. Fig. 6 shows that multi-task learning taking the intensity recognition as the auxiliary task can alleviate this phenomenon to a certain extent. Nevertheless, the multi-task learning framework trained on IEMOCAP still could not reduce the misclassification proportion of the happy emotion with weak intensity, because the corresponding samples are limited in IEMOCAP.



(a) single task learning for speech emotion recognition



(b) multi-task learning for recognition of speech emotion and intensity

Fig. 4. SER confusion matrices of single task and multi-task learning approaches on RAVDESS

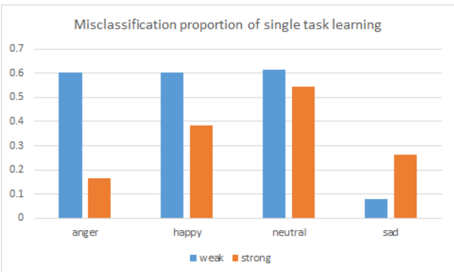


Fig. 5. Misclassification proportion of single task learning in view of each emotion and intensity on IEMOCAP

Table IV shows the comparison of our proposed multi-task framework $\mathrm{MTL}_{emo+int}$ with the existing SOTA pre-trained model in SER. The results come from [22] where the experimental setup is the same as in our work. The comparison terms are limited because the experiment in [22] was only carried out on IEMOCAP in terms of UAR. It can be observed that our model $\mathrm{MTL}_{emo+int}$ outperforms the other SOTA models and uses much fewer parameters at the same time.

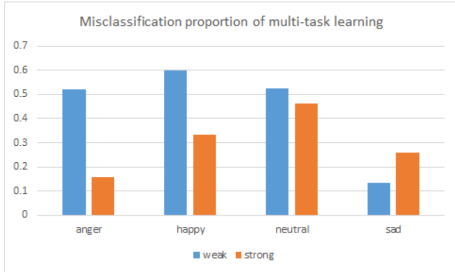


Fig. 6. Misclassification proportion of multi-task learning in view of each emotion and intensity on IEMOCAP

TABLE IV
SPEECH EMOTION RECOGNITION COMPARISON OF DIFFERENT PRE-TRAINED MODELS ON IEMOCAP

Model	UAR	Params
WavLM Large	0.7062	316.6M
HuBERT Large	0.6762	316.6M
data2vec Large	0.6631	314.3M
STL_{emo}	0.6511	96.5M
$\mathrm{MTL}_{emo+int}$	0.7082	97.3M

IV. CONCLUSIONS

Emotion intensity is important auxiliary information in emotion recognition, but its effect on emotion recognition has seldomly been studied. Inspired by the statistical relationship between emotion and intensity, we propose a multi-task learning framework that takes speech emotion recognition as the primary task and emotion intensity recognition as the auxiliary task in this paper. Experiment results show that the proposed multi-task learning framework outperforms the single task learning approach and achieves SOTA speech emotion recognition performance.

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